Find the eigenvectors of

$$\Sigma = \begin{bmatrix} 25 & 12 \\ 12 & 9 \end{bmatrix}$$

$$\det \begin{bmatrix} 25-\lambda & 12 \\ 12 & 9-\lambda \end{bmatrix} V = 0 \qquad V \neq 0$$

$$\lambda \; \frac{17 \pm 2^2\sqrt{13}}{2} \quad \text{eigenvalues}$$

$$12 x_0 + \left(\frac{1-2^2\sqrt{13}}{2}\right) x_1 = 0$$

$$12 x_0 = \frac{-1+2^2\sqrt{13}}{2} x_1$$

$$24 x_0 = (-1+2^2\sqrt{13}) x_1$$

$$V_0 = a \begin{bmatrix} -1+4\sqrt{13} \\ 24 \end{bmatrix} \qquad V_1 = b \begin{bmatrix} -1-4\sqrt{13} \\ 24 \end{bmatrix}$$

$$V = \begin{bmatrix} -1+4\sqrt{13} & -1-4\sqrt{13} \\ 24 & 24 \end{bmatrix} \cdot \frac{1}{24 \cdot 8\sqrt{13}}$$

$$\det V = (-1+4\sqrt{13})24 - 24(-1-4\sqrt{13})$$

$$= 24(\cancel{-1}+4\sqrt{13}\cancel{+1}+4\sqrt{13})$$

$$= 24 \cdot 8\sqrt{13}$$

$$\begin{array}{r} 24 \\ \times \quad 2 \\ \hline 48 \end{array}$$

$$\frac{1}{24 \cdot 8\sqrt{13}} \cdot \frac{\sqrt{13}}{\sqrt{13}} = \frac{\sqrt{13}}{24 \cdot 4 \cdot 2 \cdot 13}$$

$$V = \begin{bmatrix} \frac{-\sqrt{13}}{24 \cdot 4 \cdot 2 \cdot 13} + \frac{\cancel{4} \cdot \cancel{13}}{24 \cdot \cancel{4} \cdot 2 \cdot \cancel{13}} & \frac{-\sqrt{13}}{24 \cdot 4 \cdot 2 \cdot 13} - \frac{1}{48} \\ \frac{\sqrt{13}}{8 \cdot 13} & \frac{\sqrt{13}}{8 \cdot 13} \end{bmatrix}$$

$$V = \begin{bmatrix} \frac{-\sqrt{13}}{2496} + \frac{1}{48} & \frac{-\sqrt{13}}{2496} - \frac{1}{48} \\ \frac{\sqrt{13}}{104} & \frac{\sqrt{13}}{104} \end{bmatrix}$$

$$24 \cdot 8 \cdot 13 = 192 \cdot 13$$

$$8 \cdot 13 = 104$$

$$\begin{array}{r} 192 \\ \times \quad 13 \\ \hline 576 \\ 1920 \\ \hline 2496 \end{array}$$